

LDO Project Description

The objective of this design project is to implement an LDO circuit in 45nm technology that produces a **regulator output of 0.9v** with a supply voltage of 1.2v & a bandgap reference voltage of 0.75v.

The design specifications that need to be achieved are:

Supply Voltage	1.2 V
Reference Voltage	0.75 V
Regulated Voltage	0.9 V
Load Current	5 mA – 30 mA
Load Capacitor	200 pF
DC Gain	> 40 dB
Phase-Margin	> 60 degrees
PSRR at 1 MHz	> 30 dB
PSRR at DC	> 40 dB
Offset Std Dev (σ)	< 2 mV